

# Late Breaking Results: Opera: An Open and Efficient Platform for Data-driven Synthesis of Analog Circuits

Shikai Wang<sup>1</sup>, Yaolong Hu<sup>2</sup>, Zhiqiang Yi<sup>1</sup>, Taiyun Chi<sup>2</sup>, Weidong Cao<sup>1</sup>

<sup>1</sup>ECE Department of The George Washington University, Washington, D.C., USA

<sup>2</sup>ECE Department of Rice University, Houston, USA

{shikai.wang, zhiqiang.yi}@gwu.edu, {yh72, Taiyun.Chi}@rice.edu, weidong.cao@email.gwu.edu

**Abstract**—The front-end synthesis of analog circuits has been a long-standing challenge since the advent of integrated circuits. Many methods, ranging from conventional optimization-based techniques to emerging learning-based approaches, have been extensively explored to address this challenge. Yet, these methods are data-driven and often suffer from low design efficiency, due to their heavy reliance on time-consuming circuit simulators, which are frequently used in the synthesis loop for real-time evaluation of the evolving circuit design. In addition, benchmarking these methods is also largely unachievable due to their exclusive use of commercial semiconductor technology for evaluation. This “Late Breaking Results” introduces Opera, an open and efficient platform for the data-driven synthesis of analog circuits. Specifically, Opera develops efficient surrogate models for various circuits and integrates them into open-source OpenAI Gym-like environments to enable efficient synthesis. Case studies on exemplary circuits show that this platform can accelerate the conventional data-driven synthesis flow by up to 40×. It also enables the benchmarking of various synthesis methods with standardized environments built upon an open-source semiconductor process.

## I. INTRODUCTION

Analog circuit synthesis has been a longstanding challenge in the field of integrated circuits. The primary goal of the front-end synthesis is to automatically find device parameters to fulfill desired circuit specifications based on a given circuit topology. Over the past decades, many methods have been proposed to address this challenge, such as Genetic algorithms and Bayesian Optimization [1], [2]. Recent advances in deep learning further offer exciting opportunities to transform this task into a learning problem [3], [4].

Nonetheless, these data-driven methods have several critical limitations. **First, low design efficiency.** Nearly all prior methods [1]–[4] rely on time-consuming circuit simulators in the synthesis loop to evaluate evolving circuit designs in real time. Specifically, emerging learning-based approaches require extensive data generated by simulators to ensure the high quality of optimization agents. **Second, close-source and inaccessibility.** Despite numerous proposed methods and ongoing research, there is a notable lack of widely accepted benchmarks and standardized development environments for canonical assessments and reproducible research. Without tackling these critical issues, the sustainable and scalable advancement of analog synthesis will be significantly hindered.

This “Late Breaking Results” introduces Opera, an open and efficient platform for the data-driven synthesis of analog circuits. Opera leverages efficient surrogate models for various circuit types, integrating them into open-source OpenAI Gym-like environments to enable faster and more efficient analog circuit synthesis. We evaluate the efficacy of these surrogate models across several representative analog building blocks, demonstrating that they can accelerate the data-driven synthesis process by up to 40×. Additionally, Opera can be used to benchmark various synthesis methods with these well-developed circuit gym environments.

## II. RELATED WORK

Various design automation techniques have been proposed for the front-end synthesis of analog circuits, mainly falling into two

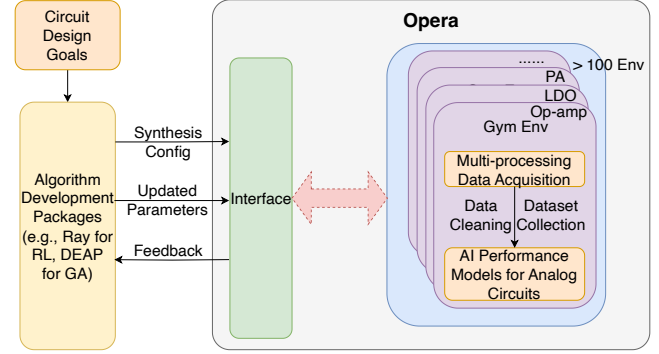


Fig. 1: An overview of the proposed Opera platform, which has multiple Circuit Gym Environments and a unified Interface. Most Gym environments are developed with the open-source 130 nm technology from SkyWater. The bidirectional interaction between the environments and the external Algorithm Development Packages enables efficient and universal circuit synthesis and benchmarking.

categories: conventional optimization-based methods and emerging learning-based methods. Optimization-based methods [1], [2] include Bayesian Optimization, Geometric Programming, Genetic Algorithms (GA), and Simulated Annealing. These methods use corresponding algorithms to search for optimal device parameters. Learning-based methods have recently emerged to address the same problem by training an agent. Supervised and reinforcement learning (RL) approaches have been widely studied. These methods [3], [4] can potentially tackle the design of large-scale analog circuits. Yet, all these methods are data-driven and require frequent interaction with time-consuming circuit simulators during the synthesis process, which leads to low design efficiency. The challenge is more pronounced for learning-based methods, which need large-volume data to ensure high training quality. In addition, nearly all prior methods use commercial semiconductor technology for evaluation, making benchmarking and reproducible research extremely difficult. Addressing these critical challenges is the key to enabling the sustainable and scalable advancement of analog synthesis.

## III. OPERA PLATFORM

The proposed Opera platform consists of two key components, multiple **Circuit Gym Environments** and **Interface Signals**, and interacts with external Algorithm Development Packages to develop/benchmark various synthesis methods, as shown in Fig. 1.

Each **Circuit Gym Environment** corresponds to a specific analog circuit design, such as operational amplifiers (Op-Amps) or low-drop voltage regulators (LDOs). At the core of each environment is a **circuit performance evaluation engine** (i.e., a learning-based surrogate model), which provides circuit performance prediction based on given input features (i.e., device parameters) almost instantaneously compared to conventional time-consuming SPICE sim-

TABLE I: R-Squared metrics for Op-Amp, LDO, and PA designs with two different surrogate models.

| Design | Metric         | MLP     | Transformer |
|--------|----------------|---------|-------------|
| Op-Amp | DC Gain        | 0.9188  | 0.9779      |
|        | Power          | 0.9243  | 0.9453      |
|        | Phase Margin   | 0.9328  | 0.9828      |
| LDO    | PSRR           | 0.9608  | 0.9963      |
|        | VOS            | 0.8425  | 0.9955      |
|        | Phase Margin   | 0.8265  | 0.9517      |
| PA     | AM-PM          | 0.98879 | >0.99       |
|        | Gain Expansion | 0.9922  |             |
|        | Op1dB          | 0.9945  |             |
|        | DE1dB          | 0.9978  |             |
|        | Power Gain     | 0.9986  |             |

ulations and electromagnetic (EM) simulations. Most environments are built with the open-source 130 nm technology from SkyWater. Each environment has an embedded data acquisition module that can automatically collect a specification dataset to build a surrogate model for each circuit. We develop a **Transformer-based** surrogate model to achieve high prediction accuracy. The proposed transformer-based architecture, which has rarely been explored in surrogate modeling for analog circuits, leverages the strengths of transformers – widely recognized for their success in sequence modeling and natural language processing. By utilizing self-attention mechanisms and multi-head attention architecture, the model effectively captures complex dependencies within the data. This design significantly enhances the network’s capability to model analog circuit performance with improved precision.

The **Interface Signals** facilitate bidirectional interaction between the Circuit Gym Environments and the external Algorithm Development Packages that are used to develop various optimization algorithms for analog synthesis. Typical Packages for algorithm development include Ray and DEAP for RL and GA optimization, which are well developed by prior works. Our platform can seamlessly integrate with these packages to develop and benchmark various synthesis methods. Users are also allowed to obtain datasets, build surrogate models, and optimize circuit parameters tailored to their unique analog circuits by adapting Opera.

#### IV. EXPERIMENTS

##### A. Data Acquisition and Performance Modeling

Opera supports various low-frequency analog circuits with SkyWater 130 nm open-source technology and several radio-frequency (RF) circuits with a confidential semiconductor process. We show experiments using representative circuits, including Op-Amps, LDOs, and RF power amplifiers (PAs). These designs were selected to evaluate the effectiveness and versatility of our proposed methodology across different analog circuit types. We collected circuit parameter-performance datasets for the three circuits, analyzing their parameter spaces and performance. Parameters include transistor dimensions, capacitors, and bias currents for Op-Amp and LDO and additional inductance for RF PA.

Two surrogate models were developed for each circuit: a baseline Multi-Layer Perceptron (MLP) and the proposed Transformer-based model. The Transformer consistently achieved higher R-squared values across all circuits. Both models demonstrated strong predictive performance with R-squared values exceeding 0.8. Notably, the Transformer-based model achieved R-squared values above 0.9 for all circuits, highlighting its superior capability in circuit surrogate modeling. These results underscore the advantages of using Transformer architectures for more accurate analog circuit performance prediction. The results of these surrogate models are summarized in Table I.

TABLE II: Comparison of optimized specifications across different designs (Op-Amp, LDO, and PA) using GA and RL methods, as well as the synthesis efficiency improvement with surrogate models.

| Design          | Spec (Unit)      | Target    | Model  | Sim.   | Speed-up |
|-----------------|------------------|-----------|--------|--------|----------|
| Op-Amp (RL)     | DCGAIN (dB)      | > 71.25   | 86.20  | 86.70  | ~ 40×    |
|                 | PM (deg)         | 60        | 59.55  | 58.08  |          |
|                 | Power ( $\mu$ W) | < 185     | 166.25 | 188.72 |          |
| LDO (GA)        | PSRR (dB)        | < -20     | -35.37 | -34.33 |          |
|                 | PM (deg)         | 60        | 60     | 61     |          |
|                 | VOS (mV)         | < 200     | 1      | 91     |          |
| PA (28GHz) (GA) | AMPM (deg)       | < 5       | 5.9    | 9.0    |          |
|                 | Gain Exp. (dB)   | < 0.5     | 0.46   | 0.43   |          |
|                 | OP1dB (dBm)      | > 18      | 19.64  | 19.31  |          |
|                 | DE1dB (%)        | optimized | 47.1   | 47.4   |          |
|                 | PG (dB)          | > 14      | 18.85  | 19.22  |          |

##### B. Synthesis and Comparison

We demonstrate the synthesis quality and efficiency with surrogate models by applying RL to Op-Amp and GA to LDO and PA. The design goals of these circuits follow prior published work. Specifically, for the PA, the optimization target was to ensure the device operates at 28 GHz, a typical frequency for 5G applications. The optimization results and corresponding targets are summarized in Table II. Opera enables automated front-end synthesis for analog and RF circuits by successfully meeting target specifications across key metrics. Moreover, it can significantly reduce synthesis time using surrogate models to replace expensive circuit-level simulations. Specifically, Opera achieves a 40× speed-up, e.g., optimizing circuits in 2.4 minutes with a surrogate model vs. 90 minutes with Ngspice, while maintaining high accuracy. Balancing speed and precision, Opera streamlines circuit design without compromising performance.

In the future, we will perform more experiments to benchmark various synthesis methods (e.g., RL, GA, and BO) with these standardized gym environments. We will also open source the platform to broadly benefit the analog design automation community.

#### V. CONCLUSION

In this work, we introduced Opera, an open and efficient platform for the data-driven synthesis of analog circuits. Opera develops efficient surrogate models for various circuit types with open-source technology and integrates them into open-source OpenAI Gym-like environments to enable efficient synthesis. Case studies on several circuits show that it can achieve up to 40× speed-up while maintaining high accuracy. **To foster further research and development, we will release Opera as an open-source framework.**

#### REFERENCES

- [1] T. Gu, J. Wang, Z. Bi, C. Yan, F. Yang, Y. Qin, T. Cui, and X. Zeng, “tSS-BO: Scalable Bayesian Optimization for Analog Circuit Sizing via Truncated Subspace Sampling,” in *2024 Design, Automation & Test in Europe Conference & Exhibition (DATE)*, Mar. 2024, pp. 1–6, iSSN: 1558-1101.
- [2] S. Zhang, F. Yang, C. Yan, D. Zhou, and X. Zeng, “An Efficient Batch-Constrained Bayesian Optimization Approach for Analog Circuit Synthesis via Multiobjective Acquisition Ensemble,” *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 41, no. 1, pp. 1–14, Jan. 2022, conference Name: IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems.
- [3] K. Settaluri, A. Haj-Ali, Q. Huang, K. Hakhamaneshi, and B. Nikolic, “AutoCkt: Deep Reinforcement Learning of Analog Circuit Designs,” in *2020 Design, Automation & Test in Europe Conference & Exhibition (DATE)*, Mar. 2020, pp. 490–495, iSSN: 1558-1101.
- [4] W. Cao, M. Benosman, X. Zhang, and R. Ma, “Domain knowledge-infused deep learning for automated analog/radio-frequency circuit parameter optimization,” in *Proceedings of the 59th ACM/IEEE Design Automation Conference*, 2022, pp. 1015–1020.